

Civil Engineering

Apply Mathematics Principles

Learner Notes

Arithmetic, ratios, percentages, mensuration, graphs, statistics, and civil engineering calculations.

Learning outcome	Focus
Apply arithmetic and algebraic operations	Applied mathematics in workplace and training contexts.
Apply mensuration in civil works	Applied mathematics in workplace and training contexts.
Interpret graphs and statistical data	Applied mathematics in workplace and training contexts.

1. Topic Overview

This topic develops mathematical skills required in Civil Engineering. Learners should show clear method, correct units, and technical interpretation.

2. Key Formulas

$$\text{Percentage} = \frac{\text{Part}}{\text{Whole}} \times 100\%$$

$$A = \frac{1}{2}(a + b)h$$

$$\bar{x} = \frac{\sum x}{n}$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

3. Worked-Example Method

Step 1: identify the given data. Step 2: select the correct formula. Step 3: substitute values carefully. Step 4: compute. Step 5: interpret the answer with correct units.

$$\text{Percentage} = \frac{\text{Part}}{\text{Whole}} \times 100\%$$

4. Practice Questions

1. Calculate cement-sand ratio quantities for a mortar mix.

2. Find the area of a trapezoidal drainage section.
3. Compute mean and range from site-test readings.